

Math 421: Homework 5

Solutions

Due: Oct 10, 2025

Problem 1: (Ross Chapter 2 7.4)

- (a) Let $x_n = \frac{\sqrt{2}}{n}$, then (x_n) is a sequence of irrational numbers but $\lim x_n = 0$, a rational number.
(b) Let $r_n = \left(1 + \frac{1}{n}\right)^n$ then (r_n) is a sequence of rational numbers but $\lim r_n = e$, an irrational number.

Problem 2: (Ross Chapter 2 8.1 c) $\lim \frac{2n-1}{3n+2} = \frac{2}{3}$

Proof. Let $\epsilon > 0$. Set $N = \frac{7}{9\epsilon} - \frac{2}{3}$. Assume $n > N$,

$$\begin{aligned} \left| \frac{2n-1}{3n+2} - \frac{2}{3} \right| &= \left| \frac{3(2n-1) - 2(3n+2)}{3(3n+2)} \right| = \left| \frac{6n-3-6n-4}{9n+6} \right| = \left| \frac{-7}{9n+6} \right| = \frac{7}{9n+6} \\ &< \frac{7}{9\left(\frac{7}{9\epsilon} - \frac{2}{3}\right) + 6} = \frac{7}{\frac{7}{\epsilon} - 6 + 6} = \frac{7}{\frac{7}{\epsilon}} = \epsilon. \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $\lim \frac{2n-1}{3n+2} = \frac{2}{3}$. □

Problem 3: (Ross Chapter 2 8.4)

Theorem. Let (t_n) be a bounded sequence and let (s_n) be a sequence such that $\lim s_n = 0$. Then $\lim(s_n t_n) = 0$.

Proof. Let $\epsilon > 0$. Let (t_n) be a bounded sequence, that is, there exists M such that $|t_n| \leq M$ for all n . Since $\lim s_n = 0$, there exists N such that $n > N$ implies $|s_n| < \frac{\epsilon}{M}$. Assume $n > N$, then

$$|s_n t_n - 0| = |s_n t_n| = |s_n| |t_n| \leq |s_n| M < \frac{\epsilon}{M} M = \epsilon.$$

So

$$\lim(s_n t_n) = 0. \quad \square$$

Problem 4: (Ross Chapter 2 8.5)

(a)

Theorem. Let $(a_n), (b_n)$ and (s_n) be sequences such that $a_n \leq s_n \leq b_n$. If $\lim a_n = \lim b_n = s$, then $\lim s_n = s$.

Proof. Let $(a_n), (b_n)$ and (s_n) be sequences such that $a_n \leq s_n \leq b_n$ and assume $\lim a_n = \lim b_n = s$. Let $\epsilon > 0$. Since $\lim a_n = s$, there exists N_1 such that for all $n > N_1$, $|a_n - s| < \epsilon$, or $s - \epsilon < a_n < s + \epsilon$. Similarly, since $\lim b_n = s$, there exists N_2 such that for all $n > N_2$, $|b_n - s| < \epsilon$, or $s - \epsilon < b_n < s + \epsilon$. Let $N = \max\{N_1, N_2\}$. Assume $n > N$, then

$$s - \epsilon < a_n \leq s_n \leq b_n < s + \epsilon$$

so that $|s_n - s| < \epsilon$. □

(b)

Theorem. Let (s_n) and (t_n) be sequences such that $|s_n| \leq t_n$ for all n . If $\lim t_n = 0$, then $\lim s_n = 0$.

Proof. Let (s_n) and (t_n) be sequences such that $|s_n| \leq t_n$ for all n . Then $-t_n \leq s_n \leq t_n$. Let $\epsilon > 0$. Since $\lim t_n = 0$, there exists N such that if $n > N$, $|t_n| < \epsilon$. Assume $n > N$, then

$$|-t_n - 0| = |-t_n| = |t_n| < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $\lim -t_n = 0$. Thus we satisfy the hypotheses of the Squeeze Lemma, (a), and therefore $\lim s_n = 0$. $\square$

Problem 5: (Ross Chapter 2 8.8 b) $\lim[\sqrt{n^2 + n} - n] = \frac{1}{2}$.

Proof. Let $\epsilon > 0$. Set $N = \frac{1}{4\epsilon} - \frac{1}{2}$. Assume $n > N$, then

$$\begin{aligned} \left| \sqrt{n^2 + n} - n - \frac{1}{2} \right| &= \left| \sqrt{n^2 + n} - \left(n + \frac{1}{2}\right) \right| = \left| \sqrt{n^2 + n} - \left(n + \frac{1}{2}\right) \frac{\sqrt{n^2 + n} + \left(n + \frac{1}{2}\right)}{\sqrt{n^2 + n} + \left(n + \frac{1}{2}\right)} \right| \\ &= \left| \frac{n^2 + n - \left(n + \frac{1}{2}\right)^2}{\sqrt{n^2 + n} + \left(n + \frac{1}{2}\right)} \right| = \left| \frac{n^2 + n - \left(n^2 + n + \frac{1}{4}\right)}{\sqrt{n^2 + n} + \left(n + \frac{1}{2}\right)} \right| = \left| \frac{1}{4\left(\sqrt{n^2 + n} + \left(n + \frac{1}{2}\right)\right)} \right| \\ &\leq \left| \frac{1}{4\left(n + \frac{1}{2}\right)} \right| = \left| \frac{1}{4n + 2} \right| = \frac{1}{4n + 2} < \frac{1}{4\left(\frac{1}{4\epsilon} - \frac{1}{2}\right) + 2} = \frac{1}{\frac{1}{\epsilon} - 2 + 2} = \epsilon. \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $\lim[\sqrt{n^2 + n} - n] = \frac{1}{2}$. $\square$

Problem 6: (Ross Chapter 2 8.9)

(a)

Theorem. If $s_n \geq a$ for all but finitely many n , then $\lim s_n \geq a$.

Proof. Let (s_n) be a sequence where $s_n \geq a$ for all but finitely many n . Then there exists $N_0 \in \mathbb{N}$ such that $s_n \geq a$ for all $n > N_0$. Let $s = \lim s_n$, assume, by way of contradiction, that $s < a$. Let $\epsilon = a - s > 0$ and let $N \geq N_0$ so that $|s_n - s| < \epsilon$ for $n > N$. Since $s_n \geq a > s$ for all $n > N \geq N_0$ we know $s_n - s > 0$. Therefore,

$$|s_n - s| < \epsilon = a - s \Rightarrow s_n - s < a - s \Rightarrow s_n < a$$

This is a contradiction, and so our initial assumption that $s < a$ is false and therefore $s \geq a$. $\square$

(b)

Theorem. If $s_n \leq b$ for all but finitely many n , then $\lim s_n \leq b$.

Proof. Let (s_n) be a sequence where $s_n \leq b$ for all but finitely many n . Then there exists $N_0 \in \mathbb{N}$ such that $s_n \leq b$ for all $n > N_0$. Let $s = \lim s_n$, assume, by way of contradiction, that $s > b$. Let $\epsilon = s - b > 0$ and let $N \geq N_0$ so that $|s_n - s| < \epsilon$ for $n > N$. Since $s_n \leq b < s$ for all $n > N \geq N_0$ we know $s_n - s < 0$. Therefore,

$$|s_n - s| < \epsilon = s - b \Rightarrow -(s_n - s) < s - b \Rightarrow -s_n + s < s - b \Rightarrow b < s_n$$

This is a contradiction, and so our initial assumption that $s > b$ is false and therefore $s \leq b$. $\square$

(c) Combining parts (a) and (b) we see $a \leq \lim s_n \leq b$, so that $\lim s_n \in [a, b]$.